

Aya Almalla

AI Engineer

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PROFESSIONAL SUMMARY

Building AI solutions that solve real-world challenges by turning ideas into practical, deployable products. Artificial Intelligence Engineering graduate with hands-on experience across computer vision, NLP, RAG systems, and end-to-end AI development. Experienced in driving technical workflows from data preparation and model development through evaluation, deployment, and product integration, with additional strengths in team leadership and product thinking.

EDUCATION

Damascus University

Sep 2021 – Aug 2026

B.Sc. in Information Technology Engineering – Artificial Intelligence Specialization

TECHNICAL SKILLS

Programming: Python, SQL, Dart

AI & Machine Learning: Machine Learning, Deep Learning, Dataset Curation, Data Preprocessing, Fine-Tuning, Model Evaluation, Error Analysis

Computer Vision: Image Classification, Object Detection, Segmentation, OpenCV, YOLO, U-Net++, ResNet34

NLP & Generative AI: Natural Language Processing, RAG, Hugging Face Transformers, Sentence-Transformers

ML Frameworks & Data Libraries: PyTorch, TensorFlow, NumPy, Pandas

AI Engineering, MLOps & Deployment: FastAPI, REST APIs, Inference Pipelines, Model Deployment, Docker, CI/CD, MLflow, Modal, Streamlit

Databases & Data Systems: PostgreSQL, ChromaDB, Kafka, Firebase

Development & Software Engineering: Git, GitHub, Object-Oriented Programming, Architectural Design Patterns, Flutter, BLoC/Cubit

Soft Skills: Analytical Problem Solving, Team Leadership, Cross-Functional Collaboration, Technical Communication, Ownership, Product Thinking

PROFESSIONAL EXPERIENCE

AI / Computer Vision Trainee & Team Leader | EyeAI, MozaicAI – Remote

Apr 2026 – Aug 2026

- Led the project team and AI workstream, overseeing retinal dataset cleaning and balancing, model selection, experimentation, and evaluation.
- Trained and evaluated YOLO26 for five-grade diabetic retinopathy classification, achieving a 0.93 Macro AUC on the test set, and assessed U-Net++ with a ResNet34 encoder for retinal segmentation.
- Conducted a Syrian market and go-to-market analysis informed by discussions with two ophthalmologists, covering target users, competitive positioning, business model, pricing, partnerships, risks, and KPIs.
- Awarded Best Leader among 8 team leaders, while the project was recognized as Best Team among 8 teams.

Contract Computer Vision Engineer | Medical AI – Al-Khatib Lab

Mar 2025 – Sep 2025

- Contributed to HaloScan, an AI-powered Antibiotic Susceptibility Testing system built on 320 real laboratory images from Al-Khatib Lab in Damascus.
- Labeled the YOLO dataset, trained the antibiotic-disc detection model, and supported the research and evaluation workflow.
- The pipeline achieved 0.90+ precision for antibiotic-disc detection, 0.94 Dice, and 0.91 IoU for inhibition-zone segmentation.
- Contributed to reducing the end-to-end analysis and reporting workflow from 10–15 minutes manually to approximately 1–2 minutes.

PROJECTS

CADify – Graduation Project | AI-Powered Floor Plan to CAD & 3D Conversion

Tech: Python, PyTorch, Computer Vision, Segmentation, OpenCV, FastAPI, Modal

- Engineered an AI pipeline that converts raster architectural floor plans into editable DXF/CAD files compatible with AutoCAD workflows, with support for 3D scene generation.
- Designed and implemented the end-to-end topology reconstruction stage, covering wall connectivity, junctions, openings, room boundaries, structured JSON handoff, evaluation, and downstream CAD/3D integration.
- Evaluated the pipeline against 15 manually labeled floor plans, achieving 0.93 F1 for wall reconstruction and 0.99 F1 / 0.96 IoU for door and window reconstruction.
- Deployed the inference pipeline through FastAPI and Modal, reducing a typically 3-hour manual conversion workflow to approximately 1–2 minutes per floor plan.

Shifaa – Arabic Medical Question Classification

Tech: Python, PyTorch, Hugging Face Transformers, AraBERT, Streamlit

- Developed an end-to-end Arabic NLP pipeline for 16 medical specialties, performing EDA and data-leakage analysis while benchmarking TF-IDF/traditional ML, CNN/LSTM, and AraBERT approaches.
- AraBERT achieved 0.82 accuracy and 0.70 macro-F1; deployed the selected model on Hugging Face and integrated it into an interactive Streamlit application.

Lecture-Saver – RAG PDF Assistant

Tech: Python, ChromaDB, Sentence-Transformers, Gemini API, Streamlit

- Implemented a RAG assistant for course PDFs using semantic retrieval and an LLM, covering document ingestion, chunking, embedding-based retrieval, metadata filtering, and file/page-level source citations.

Equipment Utilization & Activity Classification Prototype

Tech: Python, YOLOv8, OpenCV, Kafka, PostgreSQL, Streamlit, Docker

- Developed a video-based AI system for tracking excavator activity and automatically generating analytics for active time, idle time, utilization rate, and machine activity states.

ADDITIONAL PROJECTS

- Automated Zuma Player – Developed a real-time classical computer vision automation system using motion, circle, and color analysis.
- Antibiotic Expert System – Designed a rule-based medical decision-support prototype.
- Voyago – Developed a Flutter travel application supporting trip customization, reservations, payments, and maps.

COURSES & PROFESSIONAL DEVELOPMENT

Aga Khan Foundation – Youth Entrepreneurship and Employable Skills (YES)	Aug 2026
• Completed entrepreneurship and product-development training covering Business Model Canvas (BMC), market understanding, opportunity validation, and turning ideas into executable business concepts.	
Global Entrepreneurship Week Syria – Startup Syria, Entrepreneurship Bootcamp	Jan 2026
• Completed a three-day intensive program focused on startup idea development, market positioning, and marketing strategy.	
Machine Learning by Andrew Ng – Coursera	2024
Dart & Mobile Development Practices	2023 – 2024

LEADERSHIP & VOLUNTEERING

HR Manager RBCs Team	Sep 2022 – Mar 2023
• Coordinated onboarding, training, progress tracking, and ongoing support for 40 trainees and their team leads, collaborating with Finance and IT departments.	
• Supported new-member selection through surveys and evaluation, tracked individual and team development, and helped resolve operational and interpersonal challenges.	
Content Writer PTC at Damascus University	Sep 2023 – 2024
• Created educational and technical content supporting student learning and community engagement.	
Organizing Team Member Projects Exhibition	2024
• Supported event organization and coordination for the faculty projects exhibition.	

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency